1.What is a null hypothesis (H_0) and why is it important in hypothesis testing?

- The null hypothesis (H_0) is a statement in statistics that assumes there is no significant effect, difference, or relationship between variables in a population. It is the initial assumption that researchers test using sample data.

Example

Suppose a company claims that its employees work an average of 8 hours per day.

- Null hypothesis (H_0): The average working time is 8 hours per day.
- Alternative hypothesis (H_1): The average working time is not 8 hours per day.

Mathematically:

$$H_0: \mu = 8$$

$$H_1: \mu \neq 8$$

Here, μ represents the population's average working time.

1. Provides a starting point: It establishes the initial assumption for statistical testing.
2. Helps make decisions: It helps researchers decide whether sample evidence is strong enough to reject the initial assumption.
3. Tests statistical significance: It helps determine whether an observed result could reasonably be explained by random variation.
4. Reduces bias: It provides a systematic method for evaluating claims using data.
5. Supports evidence-based conclusions: It helps researchers assess whether differences or effects are statistically significant.

2.: What does the significance level (α) represent in hypothesis testing?

-The significance level (α) is the probability of rejecting the null hypothesis (H_0) when it is actually true. In other words, it is the maximum acceptable probability of making a Type I error.

Common Significance Levels

- $\alpha = 0.10$ (10%) — A 10% risk of a Type I error.
- $\alpha = 0.05$ (5%) — The most commonly used significance level.
- $\alpha = 0.01$ (1%) — A stricter significance level requiring stronger evidence.

Example

Suppose a researcher wants to test whether a new teaching method improves students' scores.

- Null hypothesis (H_0): The teaching method does not improve scores.
- Alternative hypothesis (H_1): The teaching method improves scores.
- Significance level: $\alpha = 0.05$.

The researcher accepts a 5% risk of rejecting the null hypothesis when it is actually true.

Compare the p-value with the significance level:

- If p-value ≤ α, reject the null hypothesis.
- If p-value > α, fail to reject the null hypothesis.

3.Differentiate between Type I and Type II errors

- In hypothesis testing, Type I and Type II errors are two types of mistakes that can occur when making decisions about a null hypothesis (H_0).

1. Type I Error (False Positive)

A Type I error occurs when we reject the null hypothesis even though it is actually true.

- Probability: α (significance level)
- Example: A medical test concludes that a healthy person has a disease.

2. Type II Error (False Negative)

A Type II error occurs when we fail to reject the null hypothesis even though it is actually false.

- Probability: β
- Example: A medical test concludes that a person does not have a disease when the person actually does.

Comparison Table

Basis	Type I Error	Type II Error
Meaning	Rejecting a true null hypothesis	Failing to reject a false null hypothesis

Also called	False positive	False negative
Probability	α	β
Result	Detects an effect that does not exist	Fails to detect an actual effect
Example	Declaring an innocent person guilty	Declaring a guilty person not guilty

Easy Way to Remember

- Type I: Seeing something that is not there.
- Type II: Missing something that is actually there.

4.Explain the difference between a one-tailed and two-tailed test. Give an example of each.

- In hypothesis testing, a one-tailed test checks for an effect in one specific direction, while a two-tailed test checks for an effect in either direction.

1. One-Tailed Test

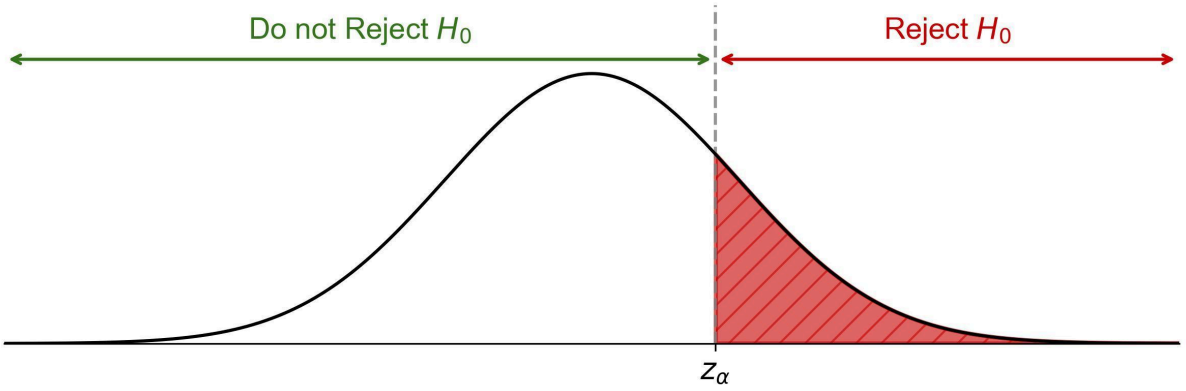
A one-tailed test is used when we want to determine whether a population parameter is greater than or less than a specified value.

Example: A company claims that its average product weight is 500 grams. A researcher wants to test whether the average weight is greater than 500 grams.

- Null hypothesis (H_0): $\mu = 500$
- Alternative hypothesis (H_1): $\mu > 500$

Here, the test looks only for an increase, so it is a right-tailed test.

Right-tailed test



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The rejection region lies in the right tail.

2. Two-Tailed Test

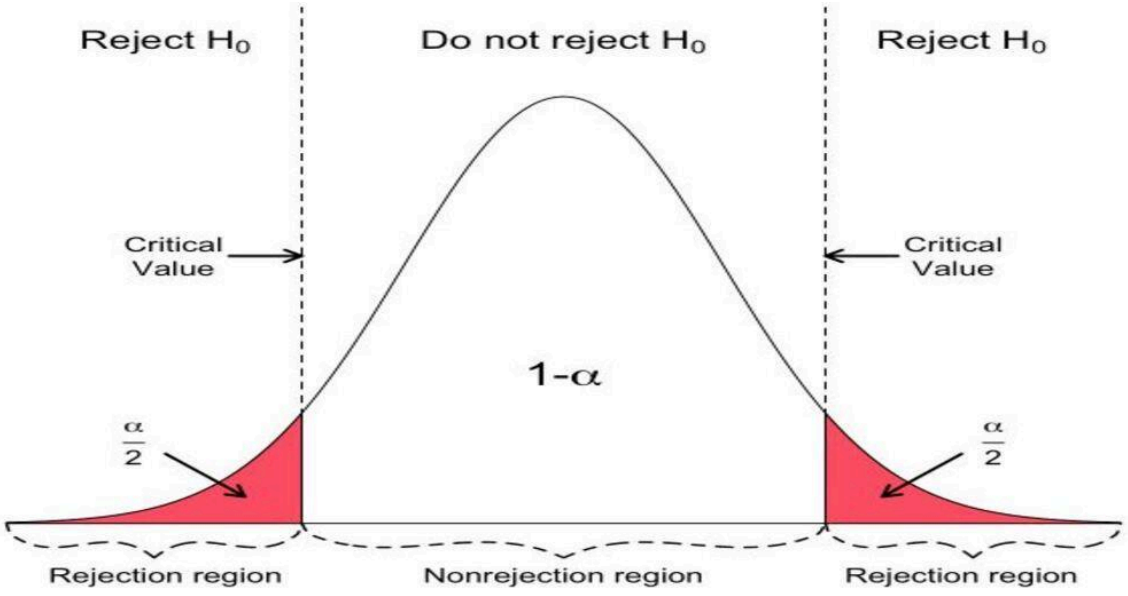
A two-tailed test is used when we want to determine whether a population parameter is different from a specified value, in either direction.

Example: A company claims that its average product weight is 500 grams. A researcher wants to test whether the average weight is different from 500 grams.

- Null hypothesis (H_0): $\mu = 500$
- Alternative hypothesis (H_1): $\mu \neq 500$

Here, the test checks for both an increase and a decrease, so it is a two-tailed test.

Two-tailed test



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The rejection regions lie in both tails.

Comparison Table

Basis	One-Tailed Test	Two-Tailed Test
Direction	One direction only	Both directions
Alternative hypothesis	$\mu > 500$ or $\mu < 500$	$\mu \neq 500$
Rejection region	One tail	Two tails
Significance level ($\alpha = 0.05$)	5% in one tail	2.5% in each tail
Example	Testing if weight is greater than 500 g	Testing if weight differs from 500 G

5.A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At α = 0.05, test the claim

- Hypothesis Testing: Customer Complaint Resolution Time

Given:

- Claimed population mean: $\mu_0 = 10$ minutes
- Sample mean: $\bar{x} = 12$ minutes
- Sample standard deviation: $s = 3$ minutes
- Sample size: $n = 9$
- Significance level: $\alpha = 0.05$

Since the population standard deviation is unknown and the sample is small, we use a one-sample t-test.

HYPOTHESIS TESTING

Step 1: State the hypotheses

Assuming we want to test whether the average resolution time differs from 10 minutes:

- Null hypothesis (H_0): The average resolution time is 10 minutes.
- Alternative hypothesis (H_1): The average resolution time is not 10 minutes.

$$H_0: \mu = 10$$

$$H_1: \mu \neq 10$$

This is a two-tailed test.

Step 2: Calculate the test statistic

The formula for the t-statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Substitute the given values:

$$t = \frac{12-10}{3/\sqrt{9}}$$

\[t=\frac{2}{3/3}=\boxed{2.00} \]

Step 3: Find the critical value

Degrees of freedom:

$$df = n - 1 = 9 - 1 = 8$$

At $\alpha = 0.05$ for a two-tailed test with 8 degrees of freedom, the critical t-value is:

$$t_{critical} = \pm 2.306$$

Step 4: Make the decision

Compare the calculated t-value with the critical value:

- Calculated t-value = 2.00
- Critical values = $\pm\ 2.306$

Since 2.00 lies between $-\ 2.306$ and $+\ 2.306$, we fail to reject the null hypothesis.

Equivalently, the p-value is approximately 0.0805, which is greater than 0.05.

Step 5: Conclusion

At the 5% significance level, there is insufficient statistical evidence to conclude that the average complaint resolution time differs from 10 minutes.

Although the sample average is 12 minutes, the difference is not statistically significant at the 5% level.

6.When should you use a Z-test instead of a t-test?

- A Z-test is used to test hypotheses about a population mean or proportion when the required assumptions are met, especially when the population standard deviation is known. A t-test is generally used to test a population mean when the population standard deviation is unknown and must be estimated from the sample.

Key Differences

Basis	Z-Test	t-Test
Population standard deviation	Known	Usually unknown
Sample size	Often large, but can be small if assumptions are met	Commonly used for small samples
Distribution	Standard normal (Z) distribution	Student's t-distribution

Shape of distribution	Fixed	Depends on degrees of freedom
Example	Testing a mean when population SD is known	Testing a mean using sample SD

Example of a Z-Test

Suppose a company claims that the average weight of its products is 500 grams. The population standard deviation is known to be 20 grams, and a random sample of 100 products has a mean weight of 505 grams.

Given:

- Population mean: $\mu_0 = 500$
- Population standard deviation: $\sigma = 20$
- Sample mean: $\bar{x} = 505$
- Sample size: $n = 100$

The Z-test formula is:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Substituting the values:

\[Z=\frac{505-500}{20/\sqrt{100}} =\frac{5}{2}=\boxed{2.5} \]

The calculated Z-value is 2.5.

When Should You Choose a Z-Test?

Use a Z-test when:

1. The population standard deviation is known for a test of the population mean.
2. You are testing population proportions under appropriate sample-size conditions.
3. The observations satisfy the relevant assumptions, such as random sampling and independence.